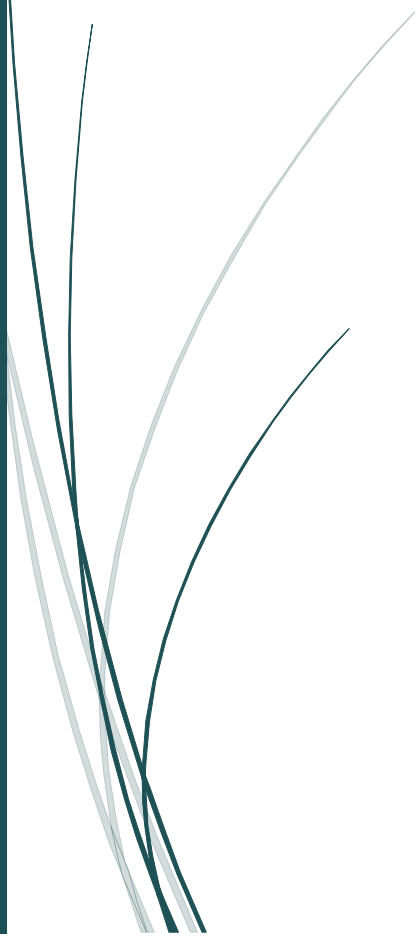
9/1/2026

N + - Network Fundamentals & Building Networks.

NAT and PAT Concepts.



YAGNIK GHETIYA

TASK 1.

- **In Cisco Packet Tracer, set up a simple network with a router, one internal PC, and one external PC. Assign private IP addresses to the internal PC and public IP addresses to the external PC, then configure static NAT on the router to allow the internal PC to communicate with the external PC.**

- **Configuring Static NAT in Cisco Packet Tracer:**

  IP Address Configuration:

Device	IP Address	Subnet Mask
Internal PC	192.168.1.10	255.255.255.0
Router (Inside)	192.168.1.1	255.255.255.0
Router (Outside)	203.0.113.1	255.255.255.0
External PC	203.0.113.2	255.255.255.0

  Configure Static NAT on Router:

  Router(config)# interface g0/0
  Router(config-if)# ip nat inside
  Router(config-if)# exit

  Router(config)# interface g0/1
  Router(config-if)# ip nat outside
  Router(config-if)# exit

  Router(config)# ip nat inside source static 192.168.1.10 203.0.113.10

  Test Connectivity: Ping the external PC from the internal PC.

  PC> ping 203.0.113.2

Result: Static NAT was configured to map the internal PC's private IP address to a public IP address, allowing communication with the external network.

TASK 2.

❖ Configure Port Address Translation (PAT) on a router in Packet Tracer so that two internal PCs with private IPs can access a simulated public web server using a single public IP address, Use the 'ip nat inside source list' command with 'overload' to enable PAT.

➢ **Configure PAT in Cisco Packet Tracer:**

IP Addressing:

- PC1: 192.168.1.10/24
- PC2: 192.168.1.20/24
- Router G0/0: 192.168.1.1/24
- Router G0/1: 203.0.113.1/24
- Web Server: 203.0.113.2/24

Router Configuration:

```
enable
configure terminal

interface g0/0
ip nat inside
no shutdown
exit

interface g0/1
ip nat outside
```

no shutdown
exit

access-list 1 permit 192.168.1.0 0.0.0.255

ip nat inside source list 1 interface g0/1 overload

Verification:

show ip nat translations

Result: Both internal PCs can access the public web server using a single public IP address through PAT.

TASK 3.

❖ Test your NAT configuration by sending a ping from the internal PC to the external PC and capture the NAT translation table on the router to verify the address mapping.

- **Test NAT Configuration:**

Ping Test:
From the internal PC, open Command Prompt and enter:

ping 203.0.113.2

Verify NAT Translation Table:
On the router, enter:

show ip nat translations

Result: The ping is successful, and the NAT translation table displays the private to public IP address mapping, confirming that NAT is working correctly.

TASK 4.

❖ Modify your Packet Tracer topology to add a third internal PC. Update your NAT or PAT configuration so all three internal PCs can access the external network, and verify that each translation appears in the router's NAT table.

➢ **Add a Third Internal PC and Update NAT/PAT:**

IP Address Configuration:

- PC1: 192.168.1.10/24
- PC2: 192.168.1.20/24
- PC3: 192.168.1.30/24
- Router G0/0: 192.168.1.1/24
- Router G0/1: 203.0.113.1/24
- External PC: 203.0.113.2/24

Update PAT Configuration:

access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface g0/1 overload

Verify NAT Translations:

Ping the external PC (203.0.113.2) from all three internal PCs.
On the router, enter:

show ip nat translations

Result: All three internal PCs can access the external network using one public IP address, and their NAT translations appear in the router's NAT table.

TASK 5.

❖ Explain in your own words how NAT and PAT help apps like Zomato or Flipkart serve millions of users with limited public IP addresses. Give two real-world scenarios where incorrect NAT/PAT setup could break connectivity for users.

- **Importance of NAT and PAT in Real-World Applications:**

  How NAT and PAT Help Apps Like Zomato and Flipkart

  NAT and PAT allow multiple users to access the internet using a limited number of public IP addresses.

  - NAT: Converts private IP addresses into public IP addresses.
  - PAT: Allows multiple users to share a single public IP address by using different port numbers.

  For example, millions of Zomato and Flipkart users can connect to online services without requiring a separate public IP address for every user.

  Two Real-World Scenarios of Incorrect NAT/PAT Configuration

  Online Payment Failure: Incorrect NAT settings may block communication between a user's device and the payment server, causing payment failures.

  Gaming or Video Calling Issues: Incorrect PAT configuration may prevent proper port mapping, causing connection failures during online gaming or video calls.

  Result: NAT and PAT help save public IP addresses and allow multiple users to access internet services efficiently.